

Environmental drivers on moth abundance

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Moths are fundamental insects to ecology, with their population trends providing an early indicator of environmental change. This project analyses long-term moth abundance data (1992-2015) from the UK Environmental Change Network (ECN) to evaluate how meteorological variables influence species-level responses, temporal dynamics and synchrony across the monitoring sites. After extensive data cleaning and initial data understanding, the three research questions were addressed using various statistic modelling techniques, from regression modelling, time-series forecasting and site correlation analysis. Ridge regression, OLS and PCA identified key environmental drivers of species sensitivity. Prophet modelling, STL decomposition and ACF/PACF analysis characterised the temporal structure of moth counts. Finally, cross-site Spearman's rank correlations and k-mean clustering identified groups of sites exhibiting synchronised activity linked to shared climatic gradients. Together, these strategies demonstrate how large-scale environmental drivers shape moth abundance patterns and highlight the value of long-term monitoring to understand ecological responses to climate variability.

1 INTRODUCTION

Moths are a significant role in the UK ecosystem, from their contribution in the food chain as prey [1] and pollinators [5] and thus are a good indicator for environmental change [2, 4]. Long term monitoring of moth populations can be used to provide insight into the ecological effects of climate variability.

This study uses long-term data from the Environmental Change Network (ECN), which is a program sponsored by multiple UK government departments and agencies focused on understanding the causes and consequences of environmental change. The data spans from 1992-2015, measuring both daily moth counts (by species) and meteorology data from 12 ECN sites across the UK. This project will help the UK Centre of Ecology & Hydrology (UKCEH) understand how biodiversity has changed under changing climate and environmental pressures.

This project investigates how meteorological variables influence moth populations at both a species and site level, using exploratory analysis, statistical modelling and time-series forecasting to identify trends and define environmental drivers.

1.1 Aims & Objectives

This project aims to assess the environmental drivers on moth abundance by answering the following research questions:

1. What moth species are the most susceptible to changes in environmental drivers?
2. What are the time series characteristics of UK moth counts? How to predict them or infill periods of missing data?
3. Are there concurrent changes in moth counts across sites? What drives them?

This will be achieved via the following objectives:

- a) **Data Cleaning & Preprocessing:** Cleaning the data, handling outliers and missing data, merging the two datasets together.
- b) **Exploratory Data Analysis (EDA):** Explore trends, seasonal patterns, variability in moth count
- c) **Environmental Driver Analysis:** Identify and compare moth species most affected by environmental changes using the ECN data and analysing the relationships between moth abundance and meteorological variables.
- d) **Time-Series Forecasting:** Develop and evaluate time-series models to characterise and predict moth count trends and evaluating them to assess their predictive performance.
- e) **Concurrent Changes Analysis:** Assessing whether concurrent changes in moth counts occur across the sites and evaluating the extent to which environmental drivers influence moth abundance.

Ridge regression, OLS and PCA identified consistent responses from moth species to precipitation, warmth and atmospheric conditions. Time-series modelling with Prophet, STL decomposition and analysis using ACF and PACF forecasted a consistent decline in moth abundance and confirmed a strong annual seasonality and short-term autocorrelation driven by weather variability. K-means clustering, followed by OLS with fixed model specifications and LASSO regression identified two clusters of sites, with temperature as the major driver of moth abundance and wind acting either as a positive or negative driver depending on location.

2 METHODOLOGY

To address our objectives, the CRISP-DM framework was used as a structured approach due to it being industry standard in data science projects [3], and for its iterative nature and clear workflow. This allowed us to ensure our research questions were addressed while also maintaining an organised approach, as follows:

- **Business & Data Understanding:** Confirming project scope and success criteria with Dr Michael Tso of the UKCEH; conducting initial data analysis to assess data quality and identify any immediate trends.
- **Data Preparation:** Cleaning and preprocessing data to create an analysis-ready dataset.
- **Modelling:** Broadly structured into two iterations; Iteration 1 focusing on addressing research questions 1 & 3 using statistical techniques; Iteration 2 focused on addressing research question 2 via time-series forecasting.
- **Evaluation:** Evaluating our findings to meet all of the objectives; answering the three research questions to develop an understanding of the effect of environmental drivers on moth abundance.

2.1 Business & Data Understanding

To directly address stakeholder needs and ensure the project was environmentally significant, Dr Tso was consulted throughout the project. Initial exploratory analysis was conducted on the raw data to assess the structure and develop an initial understanding of the relationship between moth abundance and environmental factors.

At this stage, broad seasonal patterns in moth abundance were identified, as demonstrated in figure 1.1, where moth counts are shown to consistently be higher during summer months. Annual variability was also explored, as shown by the fluctuations on figure 1.2, with 1996 demonstrating the largest moth count recorded 30,378 moth samples.

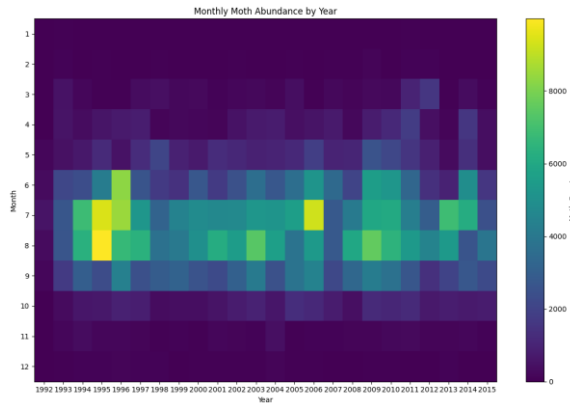


Figure 2.1: Heatmap showing monthly moth abundance by year

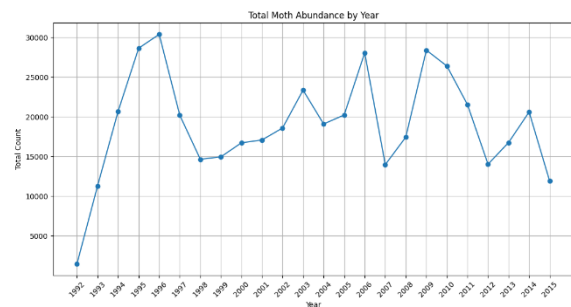


Figure 2.2: Line graph to show total moth count by year

2.2 Data Preparation

2.2.1 Moth Data

Initial data cleaning involved converting date columns to datetime64 and removing entries with FIELDNAME values of Q1 (representing quality codes in the supporting documentation for “no traps running”). There were missing values in the secondary data’s WEEK column, so a new column was created to manually calculate the value for WEEK based on the SDATE value. This needed to match the week numbers from the actual calendar year. The primary and secondary datasets for moth data were merged based on common columns so that each moth sample had data to a trap which was used for that sample.

2.2.2 Meteorological Data

All meteorological data for the year 1991 was removed due to the moth dataset recording capture from 1992 onwards. SITECODE 11 was also removed as this was not included in the moth dataset and so was redundant. SHOUR values of 24 were changed to 0 so that midnight was hour 0 as opposed to 24. Text was standardised for every measured weather variable.

Due to research question 2’s focus on time series data, the data was pivoted from long format to wide format, causing the dataset to reduce from 26 million to 2 million rows. However, some variables now had an increase of missing values after the pivot. All variables with over 20% missing values were dropped, as the imputation of such variables would introduce bias. The meteorological data was also recorded on an hourly basis as opposed to the moth data, which spanned over several days, and so was aggregated to ensure consistent data merging. The pivoted dataset was collapsed using mean for all variables apart from RAIN, resulting in a final data frame consisting of one row per day with averaged meteorological data and a daily sum for rainfall, as opposed to the initial pivot which had an hourly record.

2.2.3 Merging the datasets

In preparation for modelling and answering the research questions, the cleaned moth and weather datasets were merged into a single dataset. This required additional processing to ensure alignment and an efficient merging process.

For example, although most moth samples were recorded with a sampling duration (SPERIOD_D) of one day, others spanned multiple days or even a week. We discarded samples with sample periods exceeding a week, and expanded multiple day samples into several rows, each representing one day during which the traps were active. This was achieved by creating a new date column by generating a sequence of dates between the start of sampling (FROMDATE) and the final date (SDATE).

This was carried out with the assumption that the traps were set in the morning, so FROMDATE would represent the first full day of sampling whereas SDATE would represent the day of collection, and so was excluded from the sampling window. As a result, each sample had SPERIOD_D rows, each corresponding to one day of weather exposure.

The merging itself was carried out using shared identifiers, mainly SITECODE and the newly created date column, ensuring that each moth observation, now represented at a daily resolution, was paired to the corresponding set of meteorological data for that site on that day.

When handling outliers, the natural variability exhibited by the weather data was considered, particularly for the temperature and solar/radiation variables. To ensure we preserved extreme weather events which may have been ecologically relevant towards moth behaviour, most of the outliers were retained. However, the NETRAD variable in particular contained extreme values, which was mitigated by removing the bottom and top 1% of values, retaining 98% of the data with the most extreme 2% assumed to be a recording error. The ALGBRD variable was also dropped, due to it being measured as a w/m^2 ratio, its high frequency of extreme values and lack of interpretability within the scope of this project.

For the imputation of missing values, this was carried out using the weekly mean of each variable, which provided a stable representation of typical conditions within each week.

2.3 Modelling

2.3.1 RQ1: Identifying moth species most susceptible to environmental drivers

The aim of RQ1 was to identify which moth species demonstrated the strongest responses to environmental variation, and to determine the extent to which different meteorological variables influenced their counts.

Firstly, all weather variables were assessed for multicollinearity using a correlation matrix, as multicollinearity can inflate coefficients and obscure the interpretation of the species-specific responses. This assessment would inform later modelling choices, particularly the use of ridge regression and Principal Component Analysis (PCA).

For each species, two regression models were used to extract the coefficients of significant drivers for each species. Ridge regression was used to calculate the L2 regularisation, which would act as the penalty for multicollinearity, making it useful when considering overlapping weather variables. The biggest weather driver was then identified for the species, followed by Ordinary Least Squares (OLS), which was carried out due to its interpretability and ability to measure the size of individual effects. The p-value was extracted from the greatest driver and was deemed significant if less than or equal 0.05 because statistically a p-value less than or equal to 0.05 is recognised as significant, rejecting the null hypothesis with a 95% confidence level.

To reduce multicollinearity and identify underlying climate patterns, PCA was performed on the standardised weather variables. The principal components identified from this were then used as predictors for linear regression against each species, allowing species to be compared based on their responses to broad environmental drivers as opposed to the individual meteorological variables. The species were then ranked by their maximum sensitivity to the PCs that had the highest loading.

2.3.2 Characterising time-series structure and forecasting moth abundance

The focus of RQ2 was on understanding the temporal structure of moth abundance. Firstly, daily moth counts were aggregated to weekly totals to reduce sampling noise and provide a balance between interpretability and smoothness. To generate predictive estimates, the Prophet model was chosen due to its ability to handle strong, irregular seasonality and its ability to provide interpretable additive components. This helped predict trends and identify structural temporal patterns. STL was carried out to separate trend from seasonality, followed by ACF and PACF analysis to measure temporal dependence.

2.3.3 Detecting concurrent changes in moth counts across sites

RQ3 aimed to determine whether sites exhibit synchronized fluctuations in moth abundance, and whether these shared patterns can be explained by common environmental drivers.

Firstly, Spearman's rank correlation was calculated between all pairs of ECN sites and their changes in moth abundance, with the assumption that the relationships between the variables did not follow linear relationships, which is typical of ecological population data. This was selected due to its robustness to outliers and provided an initial indication of spatial synchrony.

To identify groups of sites exhibiting similar temporal patterns, K-Means clustering was performed, allowing for sites with similar abundance patterns to be grouped collectively. PCA was then carried out to enable plotting and see the clusters.

OLS was used as a baseline approach to establish associations between cluster-level abundance and meteorological variables. Multiple model specifications were then explored to address multicollinearity and explain variations in moth species by assigning coefficients to each weather variable based on its fluctuations and sample value.

LASSO regression was also used to further refine the identification of environmental drivers, due to its ability to identify the most influential predictors while excluding redundant or noisy variables.

Finally, PCA-based composite climate indices were constructed to summarise the major environmental gradients. These were then incorporated into confirming regression models to validate the concurrency of changes in moth counts at the synchronised sites.

3 RESULTS

3.1 What moth species are the most susceptible to changes in environmental drivers?

After constructing the correlation matrix for the weather variables shown in figure 3.1, it became clear that there was a high correlation between NETRAD and SOLAR. The matrix also demonstrated a high correlation between STMP10, STMP30, and DRYTMP, reflecting shared temperature dynamics.

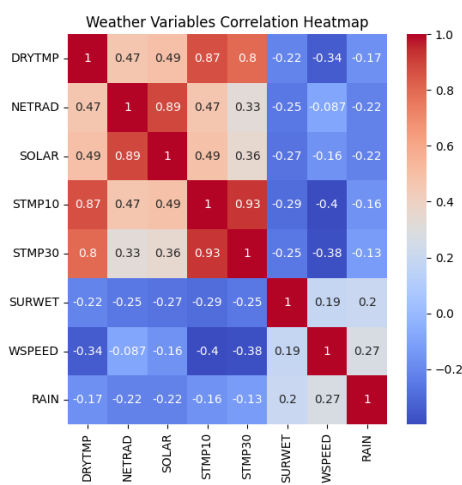


Figure 3.1: Correlation heatmap of weather variables

FIELDNAME	MAX_DRIVER	P_VALUE	COEFFICIENT	R_SQUARED
442	STMP30	5.206778e-05	1.045183	0.168869
824	STMP30	2.614652e-08	-0.915201	0.289793
537	SURWET	5.335567e-03	-0.783942	0.212373
191	SURWET	5.573536e-10	-0.637257	0.087415
461	SURWET	2.850215e-05	-0.548496	0.111963
...	...	...	...	...
2503	WSPEED	1.390369e-02	0.035138	0.066807
473	WSPEED	1.156984e-04	0.031269	0.028527
913	RAIN	8.977937e-05	0.028490	0.234118
111	RAIN	1.968307e-03	0.025850	0.193619
121	RAIN	7.058308e-04	0.020093	0.094562

Figure 3.2: Data frame of species-specific responses to environmental drivers

OLS revealed that species different not only in how strongly they responded to environmental variations, but also in which direction they responded, as shown in figure 3.2. For example, species 442 has positive coefficients, suggesting abundance increases with warmer soil temperatures. Negative coefficients, for example as shown for species 824, indicate species that decline under warmer conditions, and so could be moths that are adapted with cooler conditions, or moths that are earlier in their lifecycle and are more likely to be stressed by higher temperatures. Likewise, negative responses to SURWET suggests species that avoid wet conditions, whereas those with positive coefficients would thrive in wetter conditions. The directionally opposite responses demonstrate how moth species are highly responsive to environmental conditions and increase in abundance in conditions that they prefer.

	PC1	PC2	PC3
DRYTMP	-0.036756	0.492207	0.099026
NETRAD	-0.037591	0.270373	0.425446
SOLAR	-0.038525	0.295886	0.380841
STMP10	-0.033260	0.491914	0.045753
STMP30	-0.025758	0.439881	-0.020421
SURWET	0.031243	-0.143190	-0.045159
WSPEED	0.068315	-0.362851	0.811938
RAIN	0.994152	0.097176	-0.018867

Figure 3.3: PCA Loadings of the weather variables

FIELDNAME	PC1_SENSITIVITY	PC2_SENSITIVITY	PC3_SENSITIVITY	MAX_SENSITIVITY
2501	0.870568	-0.182324	2.694939	2.694939
208	0.124766	-1.485773	0.795572	1.485773
348	0.041418	-1.160856	-0.735014	1.160856
598	0.028704	-0.314507	1.039046	1.039046
564	0.259460	-0.851570	-0.883632	0.883632
2550	-0.010359	-0.584158	0.683689	0.683689
879	-0.600911	0.194910	-0.239858	0.600911
155	-0.051568	-0.471816	-0.528977	0.528977
475	0.000212	0.498554	0.181905	0.498554
802	0.421220	0.373250	0.281362	0.421220

Figure 3.4: Table of top ten most sensitive species and their PCA loadings

After carrying out PCA, three main components were derived, as shown in figure 3.3. PC1 was extremely dominated by RAIN with negligible contributions from the other variables, representing a pure precipitation gradient. This shows that rain varies largely regardless of the other variables, acting as a standalone environmental driver. Species 879 in figure 3.4, for example, demonstrates a heightened sensitivity to precipitation variability.

PC2 had strong positive loadings from DRYTMP, STMP10 and STMP30, with moderate contributions from SOLAR and NETRAD, which indicates it represents a temperature/energy gradient. This aligns with ecological expectations and suggests that species 208 and 348 for example would be more vulnerable to climate warming.

PC3 was dominated by WSPEED, alongside notable contribution from NETRAD and SOLAR, thus representing an atmospheric driver consisting of wind and radiation exposure. This is the maximum driver for species 598 and species 2501, the latter being the species with the highest sensitivity. This emphasises how moths are sensitive to airflow and turbulence, as it could influence their flight behaviour, but also could also play a significant role in the likelihood of trap capture.

Ultimately, the main drivers on moth abundance are precipitation, warmth and atmospheric conditions, which influence moth species in various meaningful ways.

3.2 What are the time series characteristics of UK moth counts? How to predict them or infill periods of missing data?

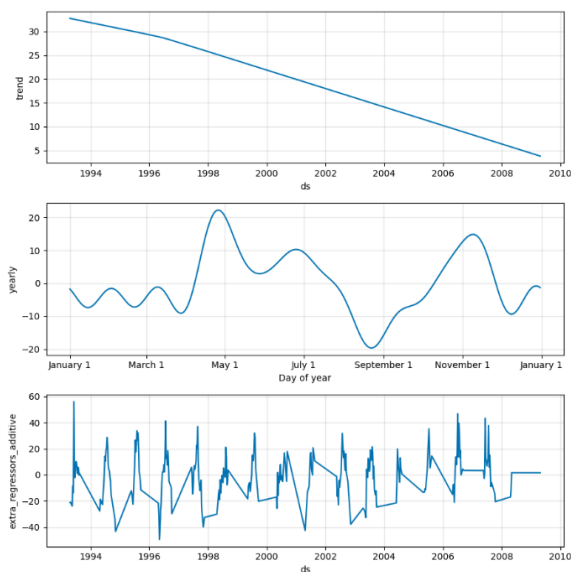


Figure 4.1: Prophet model forecasting

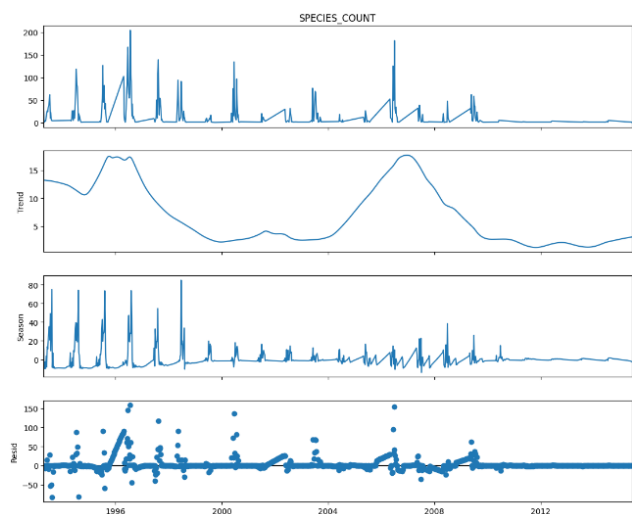


Figure 4.2: STL decomposition of seasonal structure

As shown in figure 4.1, across species the Prophet model demonstrated a forecasted decline in moth abundance, suggesting additional drivers independent of the weather variables. The yearly component captured a strong annual seasonality, with a peak in late April-May, followed by a secondary pulse in late autumn. The additive component shows frequent short-term fluctuations, with sharp peaks representing extreme weather events. This indicates that weekly variations in the weather variables strongly influence moth count and validates their use as regressors.

STL decomposition in figure 4.2 acted as an independent assessment of the temporal patterns revealed by the Prophet modelling. The trend component shows a broad rise in abundance in the late 1990s, and some medium fluctuation but no sustained directional change over the full time period. The seasonal component exhibits a highly consistent annual cycle, with sharp peaks occurring at consistent times of the year. This is reflective of the increase in abundance during summer months and reinforces the dominance of biological seasonality within temporal patterns. However, the seasonal component diminishes slightly in later years, indicating lower peak moth abundances towards the end of the time series. The residual component captures a significant year-to-year variability, showing deviations not explained by the trend nor seasonality. These fluctuations may correspond to short term environmental influences such as unusual temperature, precipitation or atmospheric conditions during the sampling period. This supports the findings from the Prophet model and provides further evidence that environmental drivers exert their influence mainly by influencing seasonal peak magnitude.

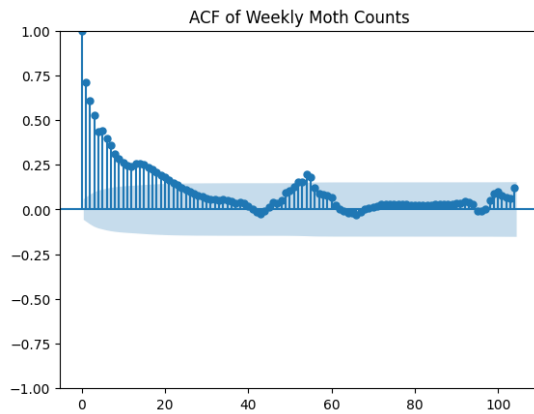


Figure 4.3: ACF plot

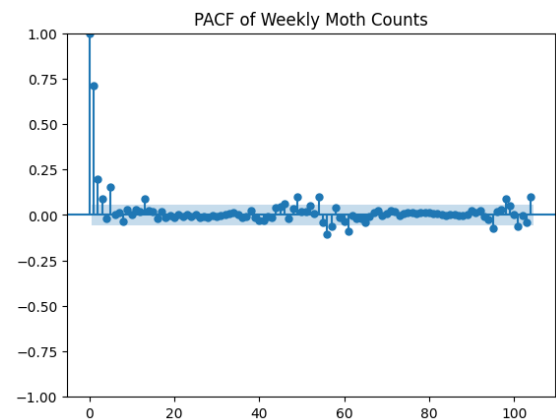


Figure 4.4: PACF plot

Beyond weather effects, the ACF plot shown in figure 4.3 demonstrates how strongly current moth counts correlate with their past values, with high autocorrelation at shorter lags indicating a strong week-to-week dependence. The secondary peak at approximately 52 weeks confirms the presence of annual seasonality. This tells us that moth populations in the UK have clear temporal memory and recurring seasonal structure.

The PACF plot in figure 4.4 shows a strong and highly significant partial autocorrelation on lag 1, which shows that moth abundance in any given week is strongly dependent on the immediately preceding week. Beyond lag 1, the partial autocorrelations drop sharply and remain close to zero, with only a small number of earlier lags showing that dependence doesn't extend far into the past, and once the influence of the previous week is accounted for, additional lags contribute little predictive information. This implies that short term processes such as weather conditions are the primary drivers of week-to-week variability.

3.3 Are there concurrent changes in moth counts across sites? What drives them?

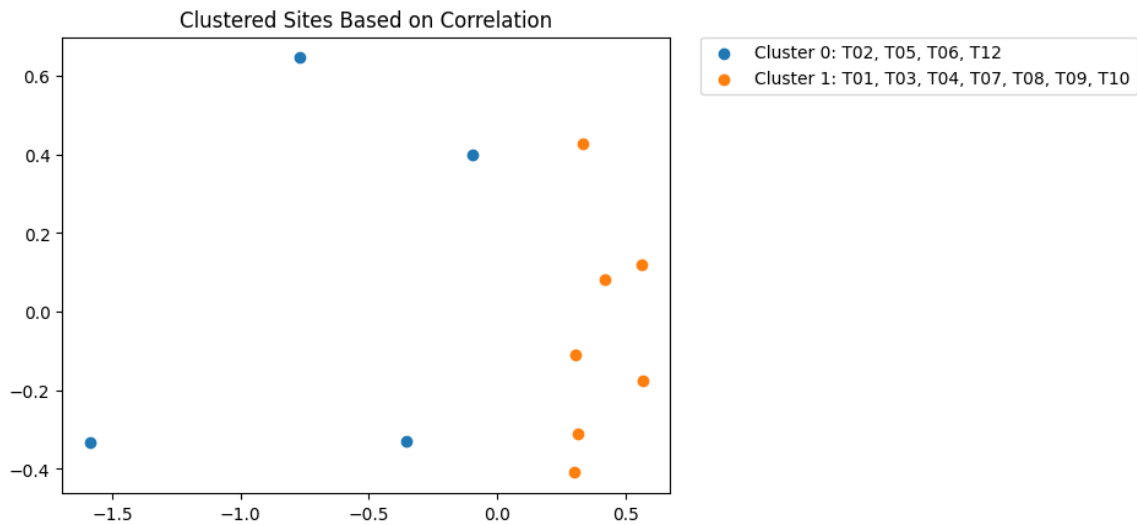


Figure 5.1: Clustered sites, with a map of positive and negative sites

K-means clustering identified two distinct clusters as shown on figure 5.1, and after OLS was applied with site-fitted effects, the component and residual plots as shown on figures 5.2 and 5.3 showed the isolated impact of each variable on clusters 0 and 1 respectively. As shown in figure 5.2, for cluster 0 temperature acted as the major positive driver, with significant support from wind, and contribution from rain and solar radiation. Figure 5.3 demonstrates temperature again as the major positive driver, however with wind acting as a negative driver. Both figures show the positive linear effect of the temperature variables consistently, with wind acting as a positive driver in some sites and a negative driver in others, depending on location. When plotted on a map as seen in figure 5.4, sites in cluster 1 can be seen as more inland whereas those in cluster 2 are more coastal, with the exception of one site in the South-East.

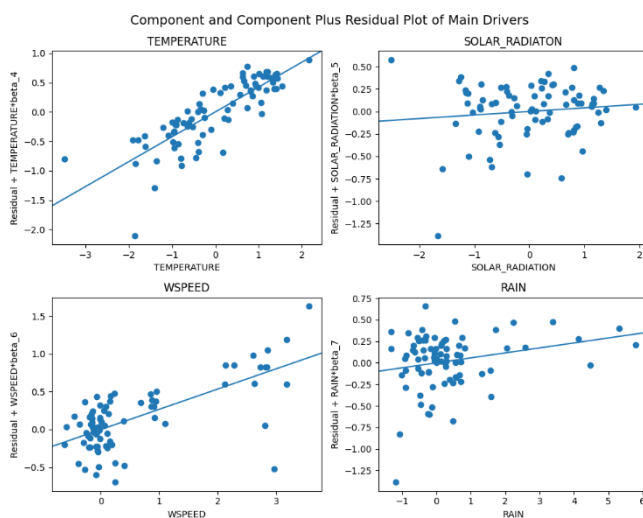


Figure 5.2: Component and residual plot of main drivers on cluster 0

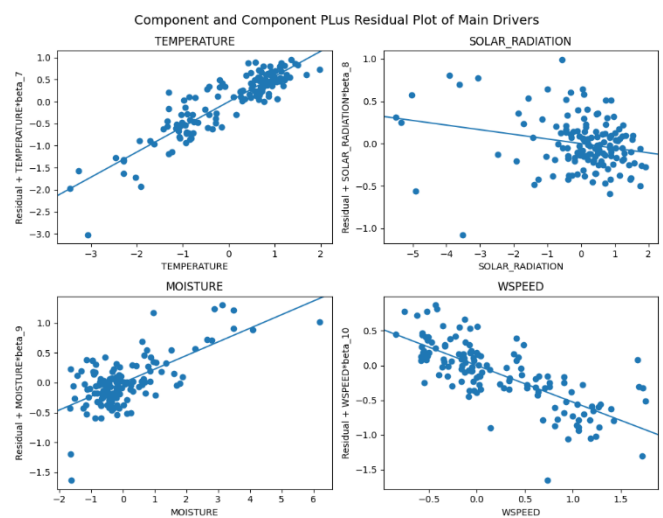


Figure 5.3: Component and residual plot of main drivers on cluster 1



Figure 5.4: Map of site locations, with green points representing cluster 0 and red representing cluster 1.

4 CONCLUSION

This project examined the effect of environmental drivers on moth abundance, influencing species-level responses, temporal dynamics, and cross-site synchrony. By using tailored approaches, all three research questions were answered.

Across species, moth abundance was strongly influenced by three environmental gradients: precipitation, temperature/energy and atmospheric conditions, and the most susceptible species were calculated. A future consideration would have been to explore machine learning techniques for regression, and research other ways to handle multicollinearity.

Temporal analysis showed that moth populations have a strong annual seasonality and short-term autocorrection, with a Prophet modelling and STL decomposition both highlighting declining seasonal amplitude in later years. These are changes in peak abundance that may reflect ecological pressures like climate change. A future consideration would have been to explore deep learning for feature abstraction and possibly investigating further what additional factors play a role outside of the weather variables.

Finally, the cross-site analysis revealed regional groups with shared weather behaviour, highlighting temperature as the main positive driver of moth abundance with wind acting either as a positive supporting driver or a negative driver.

The approach taken was appropriate given the project aims and time constraints, however future work could enhance predictive performance. It would also be interesting to expand the environmental variables further, incorporating other possible factors such as land use or light pollution to see whether they play a significant role in influencing moth abundance. As moth populations continue to decline, environmental change in the recent years is driving biodiversity loss, and so any insights into developing preventions and mitigations to protect moth populations only continues to grow more crucial.

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